

Question number	Answer	Mark
5(a)	- diameter of lesion and length of brain measured (1) - diameter of lesion multiplied by the magnification (1) Example of calculation: 6 mm and 63 mm $6 \times (150 \div 63) = 14 / 14.3 / 14.29$	(2)

Question number	Answer	Mark
5(b)(i)	- mean increase in conduction per 1 μm increase in diameter for each type of neurone calculated (1) - value for myelinated neurone divided by value for non-myelinated to give an answer that is at least 1 SF more (1) Example of calculation: myelinated = $(8.5 - 2.6) \div 2 = 2.95$ non-myelinated = $(4.2 - 2.3) \div 4 = 0.475$ $2.95 \div 0.475 = 6.2 / 6.21$	(2)

Question number	Answer	Mark
5(b)(ii)	An explanation that includes the following points: - both an increase in diameter and the presence of a myelin sheath increase speed of transmission (1) - myelin insulates the axon because it is hydrophobic (1) - therefore {nerve impulses jump from node to node / action potentials only occur at the nodes of Ranvier / saltatory conduction occurs} (1) - as diameter increases the resistance to flow decreases (1)	(4)

Question number	Answer	Additional guidance	Mark
5(b)(iii)	An answer that includes any three of the following points: • demyelination in a person with MS will result in impulses travelling slower (1) • blindness in one eye will result from demyelination of {nerve from eye to brain / one side of the brain} (1) • lack of coordination will result from sensory neurones and motor neurones transmitting nerve impulses at different speeds (1) • {blindness in one eye / lack of coordination} will result if demyelination occurs in the part of the	Accept occipital lobe	(3)